ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 1: Real Numbers (Vastavik Sankhyaye)

Key Concepts & Important Formulas:

* Euclid Division Lemma: For given positive integers a and b, there exist unique integers q and r satisfying a = bq + r, 0 <= r < b.

* Fundamental Theorem of Arithmetic: Every composite number can be expressed (factorized) as a product of primes, uniquely apart from order.

* Formula: HCF(a, b) * LCM(a, b) = a * b.

* Irrational Numbers: A number is irrational if it cannot be written in p/q form (e.g. sqrt(2), sqrt(3), sqrt(5), pi).

* Terminating vs Non-Terminating Decimals: If denominator q has prime factorization of form 2^n * 5^m, decimal expansion terminates; otherwise non-terminating repeating.

Solved Exam Examples & Practice:

Example 1: Find HCF and LCM of 12 and 18 using prime factorization.
12 = 2^2 * 3, 18 = 2 * 3^2.
HCF = 2 * 3 = 6. LCM = 2^2 * 3^2 = 36.
Verification: 6 * 36 = 216 = 12 * 18.

Example 2: Prove that sqrt(5) is an irrational number by contradiction method.